Data Cleaner and Visualiser

Product Walkthrough & Demo

A no-code platform for cleaning and visualising CSV and Excel data

data-cleaner-wine.vercel.app

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github.com/Hlulani-B/Data-Cleaner

A fast, browser-based data-cleansing and visualization workspace that bridges the gap between fragile spreadsheets and tedious custom scripts.

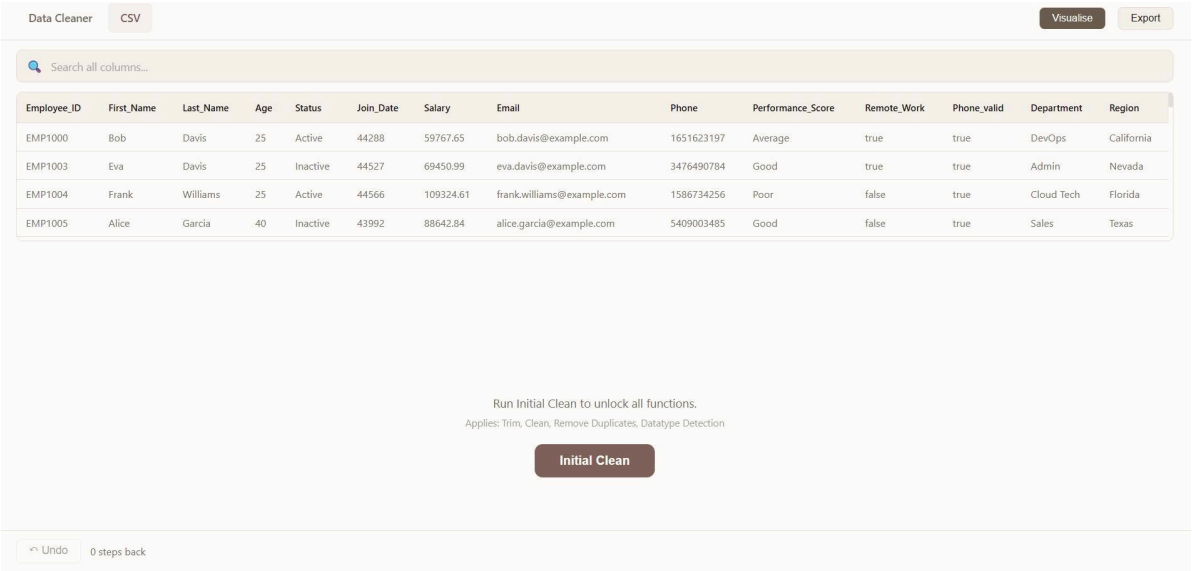
Overview

This walkthrough follows two real datasets through Data Cleaner and Visualiser, from upload to a cleaned, analysis-ready sheet. Every screenshot in this document is taken directly from the live deployed application, not a mockup, and shows the platform handling real messy data end to end: inconsistent categorical values, currency-formatted numbers, encoding artefacts, and Excel date serial numbers. Each cleaning demonstration below shows the problem as it actually appeared, the tool used to address it, and the result once applied.

The goal of the platform is simple: let anyone clean and understand messy data without writing code or wrestling with spreadsheet formulas.

Step 1 — Upload and Unlock

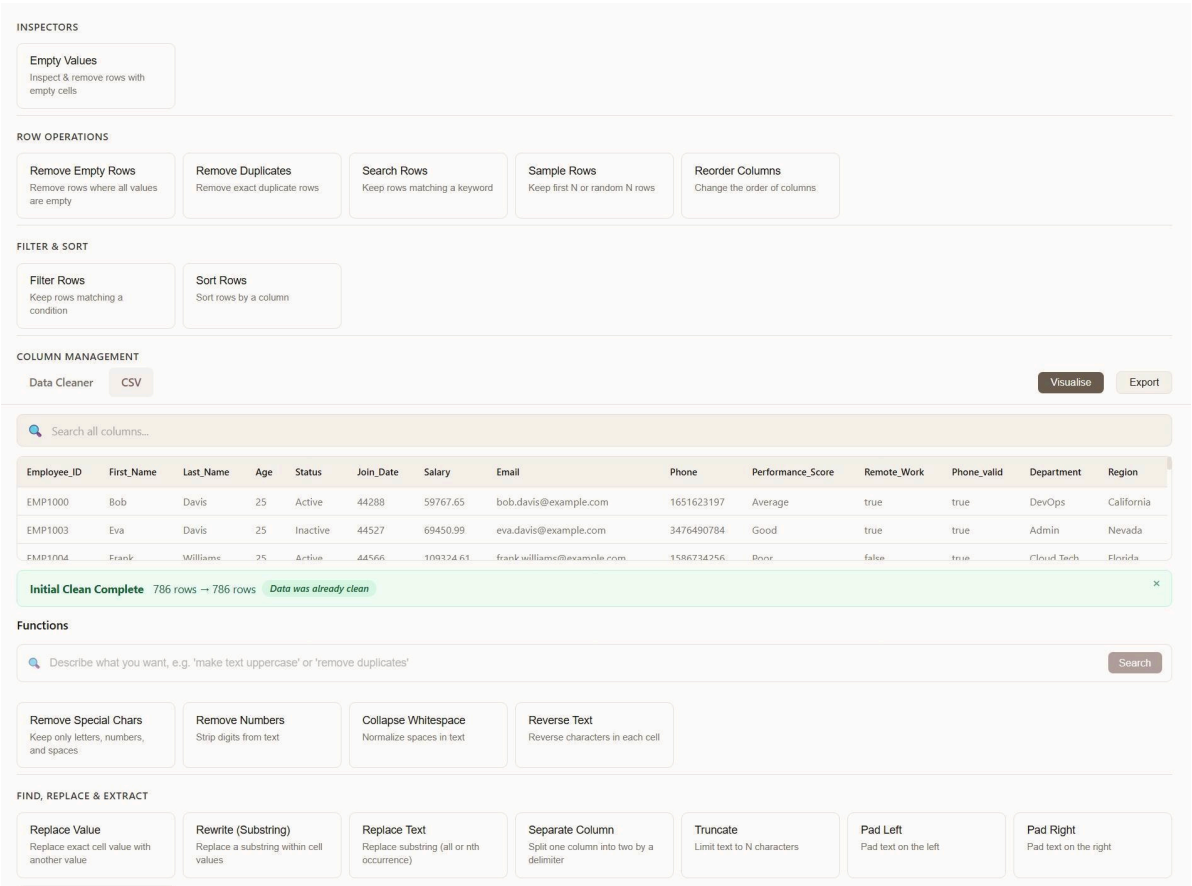
After uploading a file, the sheet loads immediately so the raw data is visible before anything is touched. All cleaning functions stay locked until the user runs the automatic Initial Clean, which trims whitespace, removes empty rows, drops exact duplicates, and detects column data types in a single action.



The employee dataset loaded and ready. The Initial Clean button unlocks the full function library.

Step 2 — Automatic Initial Clean

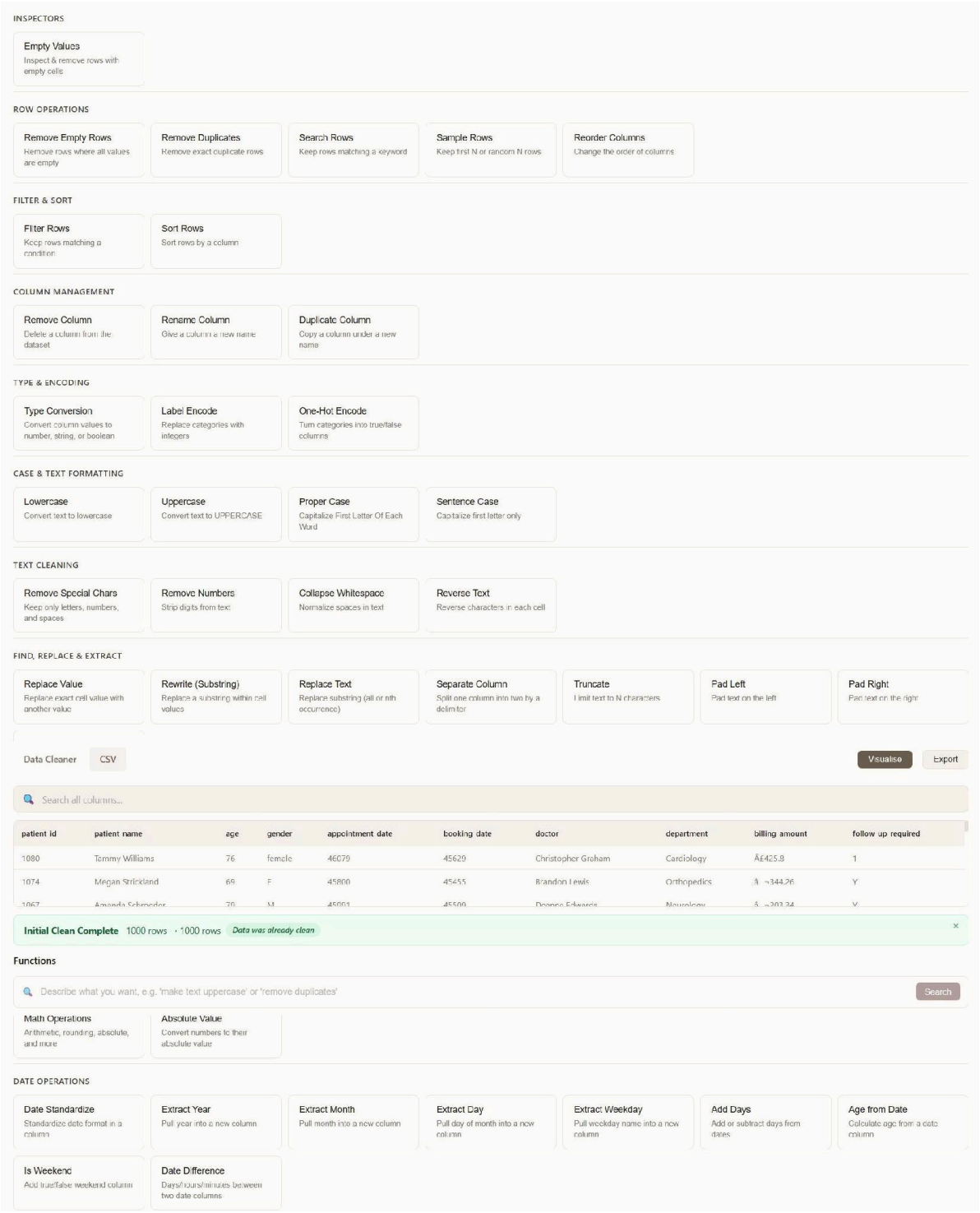
Running Initial Clean processes the whole sheet in one pass and reports exactly what changed — here, 786 rows in, 786 rows out, confirming the data was already free of empty rows or exact duplicates. Once complete, every function in the library becomes available.



Initial Clean Complete banner, with the full function library now unlocked below the sheet.

Step 3 — The Full Function Library

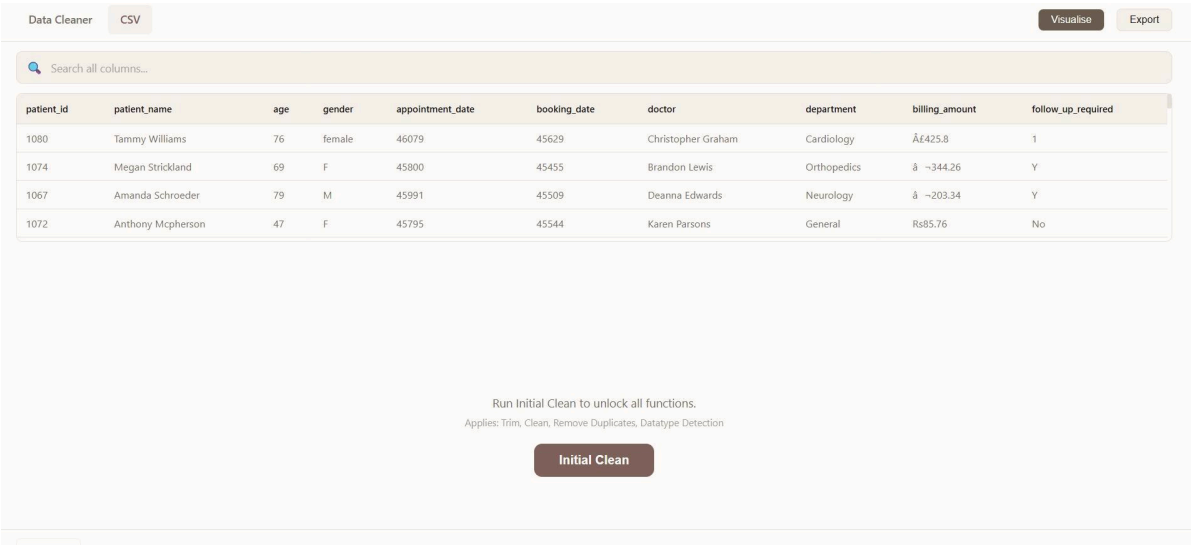
Over 40 cleaning functions are organised into clear categories so a user can scan straight to what they need: Inspectors, Row Operations, Filter & Sort, Column Management, Type & Encoding, Case & Text Formatting, Text Cleaning, Find/Replace/Extract, Math Operations, and Date Operations.



A section of the function library, covering row operations, filtering, column management, type conversion, and text cleaning.

Step 4 — A Real Messy Dataset

The rest of this walkthrough follows a second dataset, a clinic appointments sheet, before Initial Clean is run. It captures several real-world problems at once: gender recorded inconsistently, currency-formatted billing values, dates stored as raw Excel serial numbers, and a yes/no field written six different ways. The next few sections work through each of these one at a time.

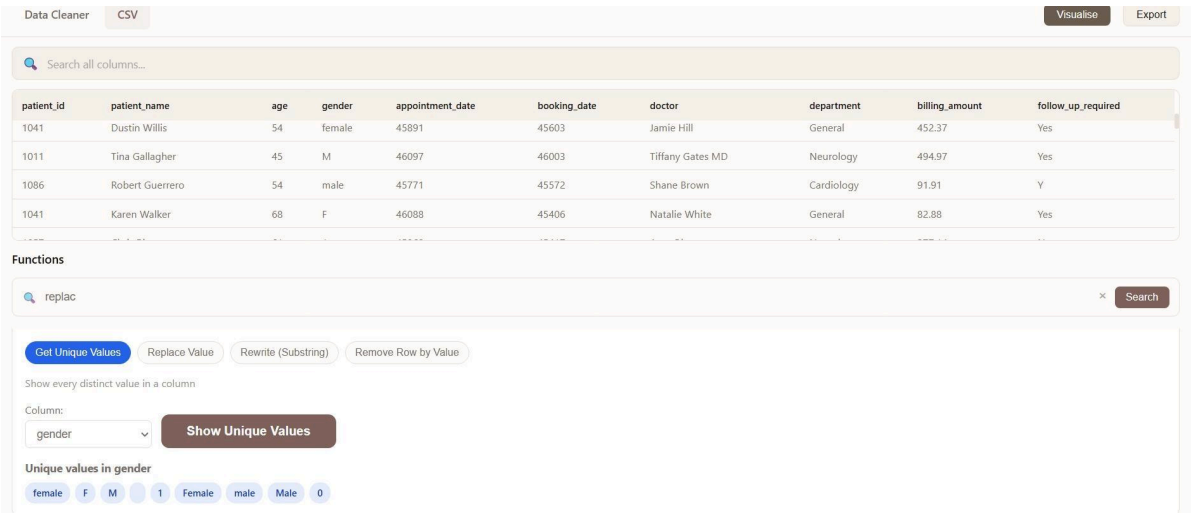


The raw appointments dataset: gender, billing_amount, appointment_date, and follow_up_required all need attention.

Step 5 — Cleaning the Gender Column

Problem

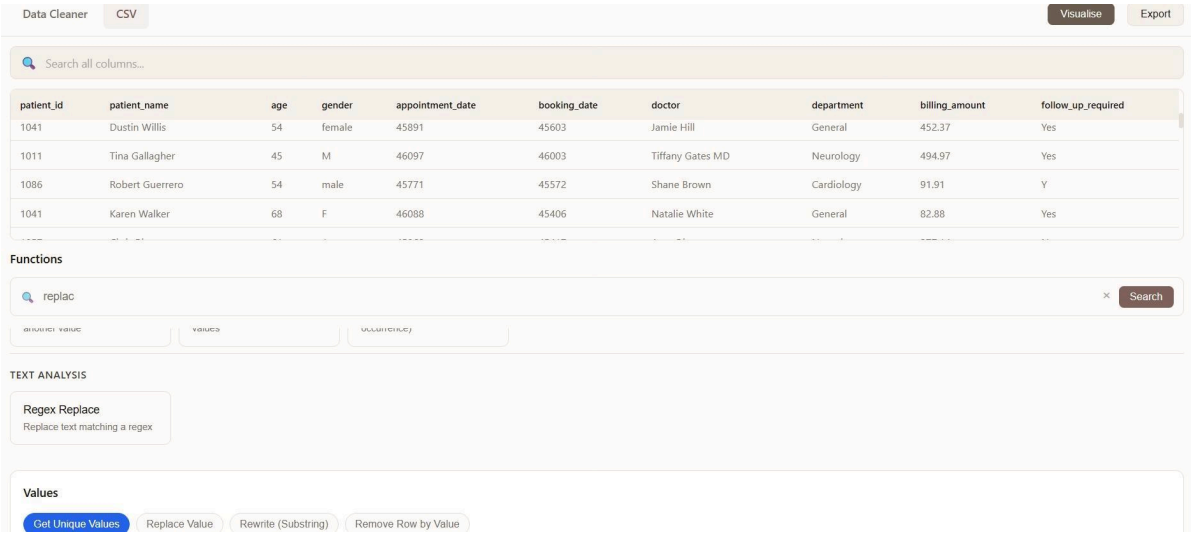
Get Unique Values on the gender column immediately surfaces the inconsistency: female, F, M, an empty value, the number 1, Female, male, Male, and 0 are all present as separate values, even though most of them mean the same thing.



Get Unique Values reveals nine different spellings and encodings of just two real categories.

Finding the right tool

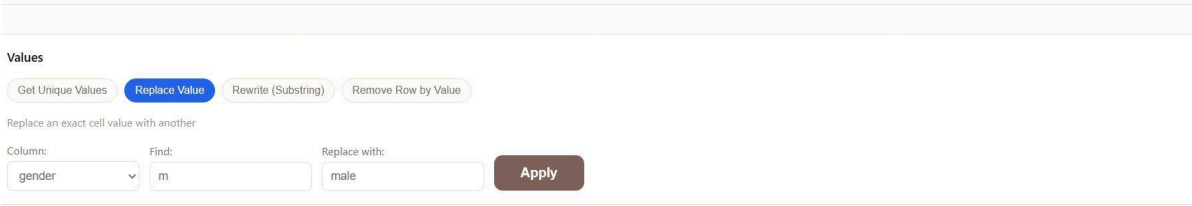
Searching the function library for 'replac' surfaces every matching tool at once, including Replace Value and Regex Replace, while the table above still shows the raw M/F values.



Searching 'replac' surfaces Replace Value and Regex Replace, with the gender column still showing raw M/F values above.

Action

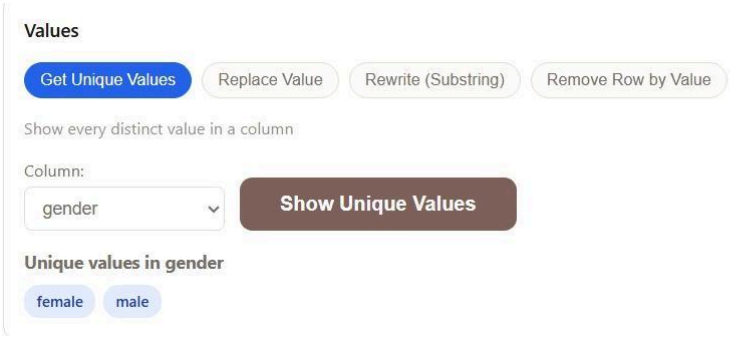
Replace Value maps an exact value to the correct one — here, replacing every 'm' in the gender column with 'male' — without writing a single formula or line of code.



Replace Value: find an exact cell value, replace it with the correct one, apply across the whole column.

Result

Running Get Unique Values again confirms the fix: gender now contains exactly two clean values.

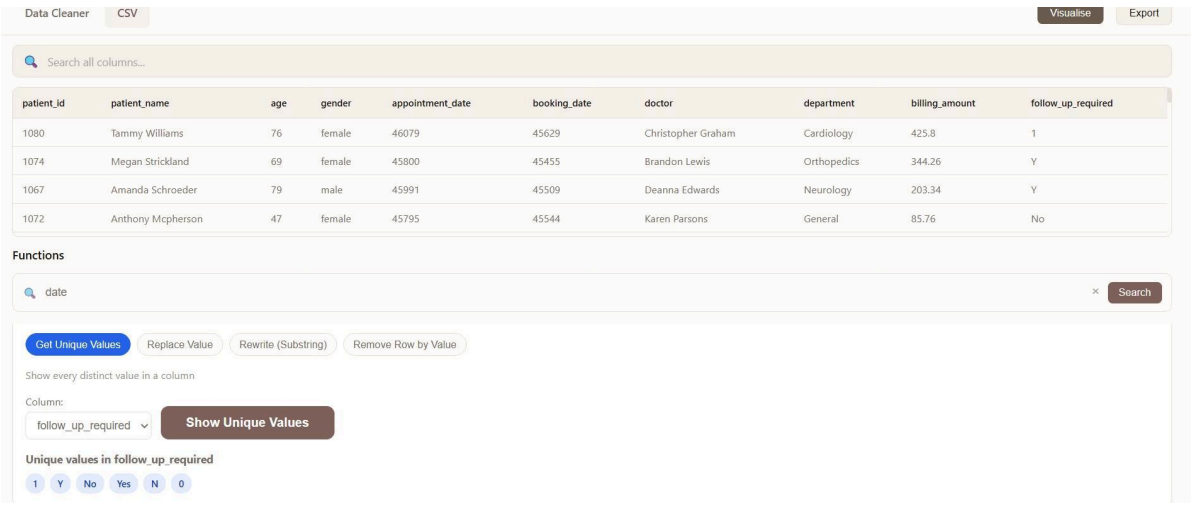


gender now shows only female and male, down from nine inconsistent variants.

Step 6 — Cleaning follow_up_required

Problem

The same kind of inconsistency shows up in follow_up_required. Viewed in full context, the column mixes 1, Y, No, Yes, N, and 0 to represent what should be a simple yes/no flag.



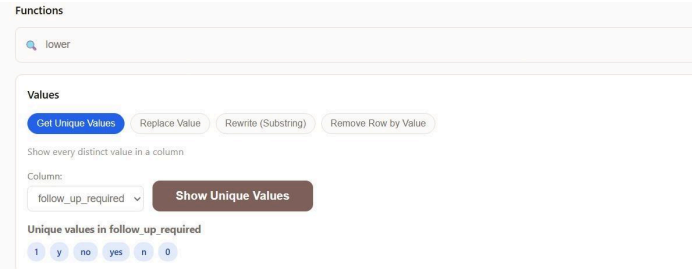
follow_up_required mixes six different representations of the same yes/no answer.

Zoomed in on just the column, the inconsistency is even easier to spot at a glance.



A close-up of follow_up_required: 1, Y, Y, No, all meaning yes or no.

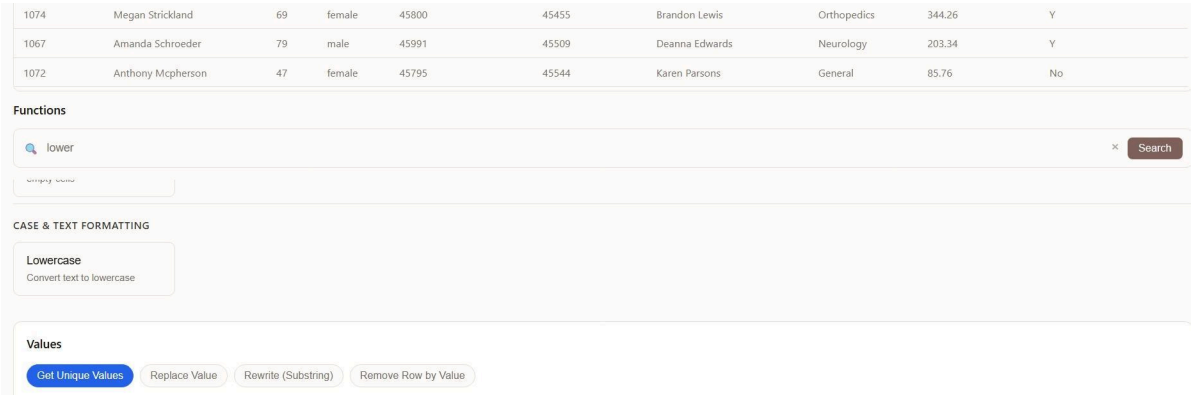
The same check on a different data load shows the lowercase variants too — y and n alongside the mix of 1, 0, Yes, and No.



Six different representations of a binary yes/no value, caught before they reach a visualisation.

Action

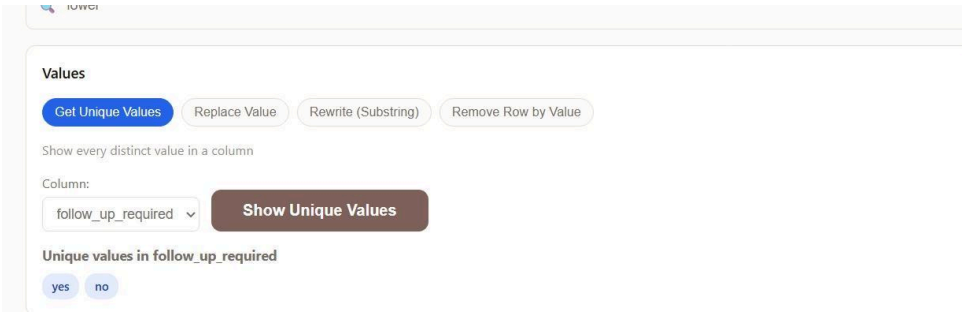
Searching 'lower' finds the Lowercase function, ready to normalise casing before the values are consolidated further — the table above still shows the raw Y/No mix.



Searching 'lower' surfaces the Lowercase function, with follow_up_required still showing Y and No above.

Result

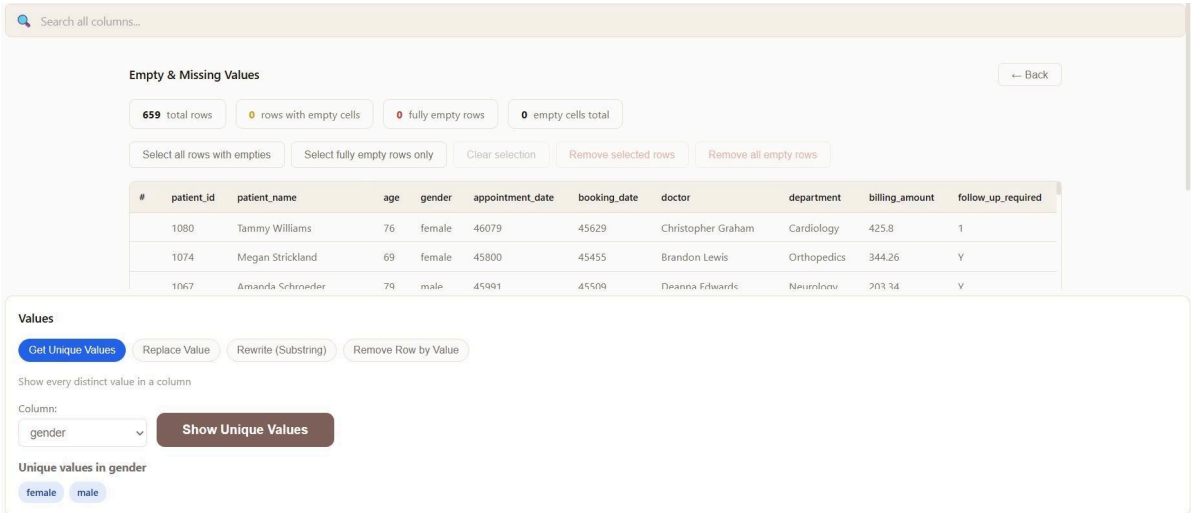
After cleaning, Get Unique Values confirms follow_up_required now holds exactly two consistent values.



follow_up_required now reads as a clean yes/no field.

Step 7 — Empty Values Inspector

The Empty Values Inspector gives a dedicated view for missing data: total rows, rows with empty cells, fully empty rows, and total empty cells, with colour-coded highlighting so problem rows are easy to find and remove in bulk.

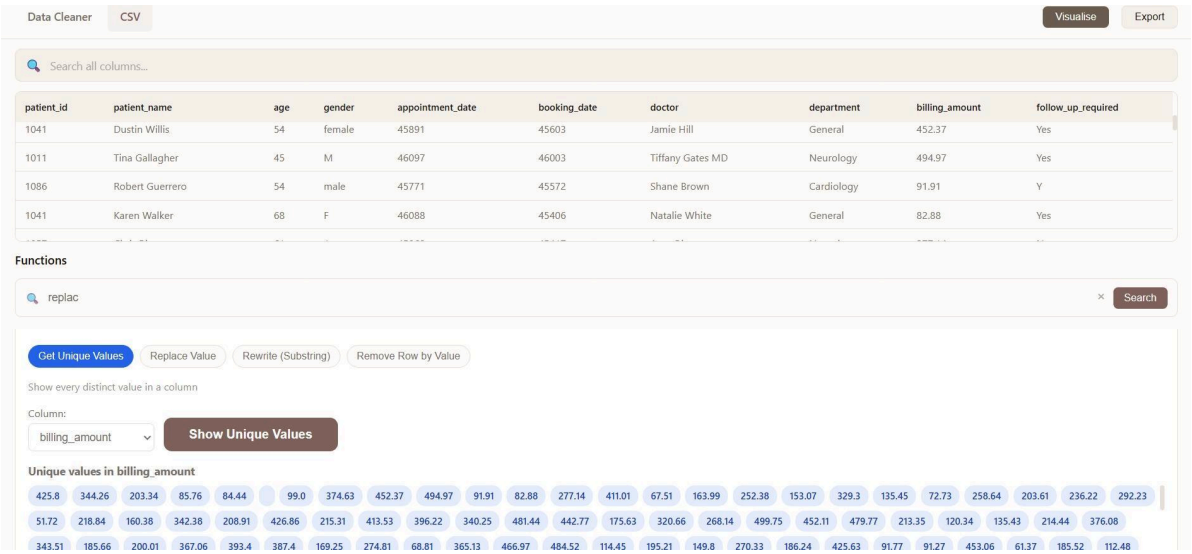


659 total rows inspected, with a clean bill of health on empty cells for this dataset.

Step 8 — Repairing Corrupted Currency Text

Baseline

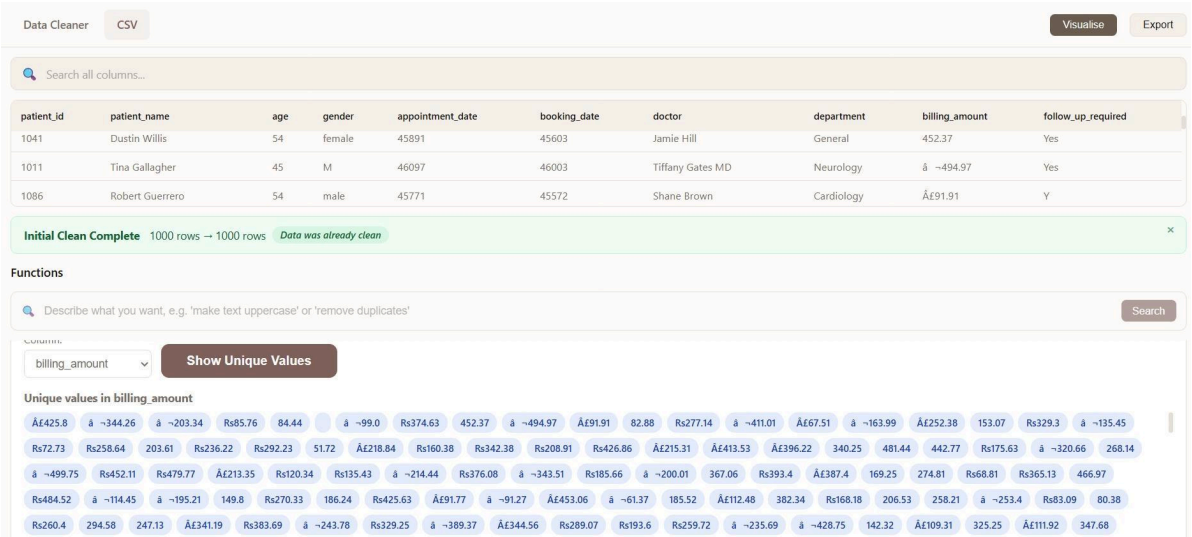
Before the encoding issue appears, Get Unique Values on billing_amount shows exactly what the column should look like: plain numeric values with no formatting attached.



billing_amount as plain numbers, before any currency-symbol corruption is introduced.

Problem

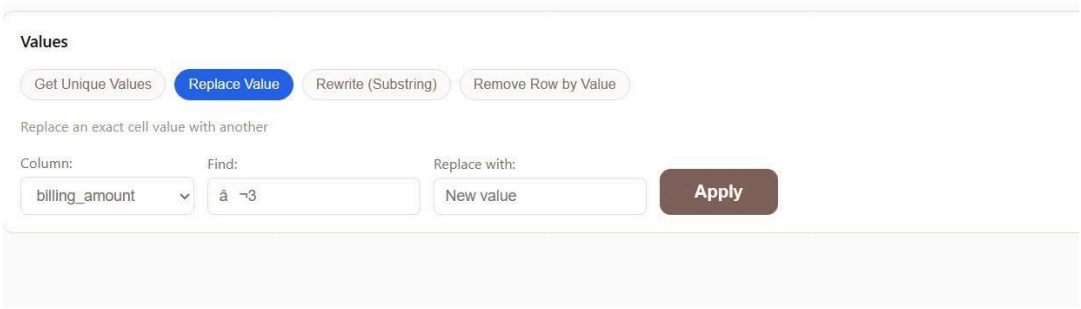
On this dataset, currency symbols copied from a source spreadsheet were encoded as UTF-8 but read back incorrectly, producing garbled prefixes such as 'â ¬', 'Â£', and 'Rs' scattered across the same column.



billing_amount mixing three different corrupted currency prefixes across otherwise identical values.

First approach: Replace Value

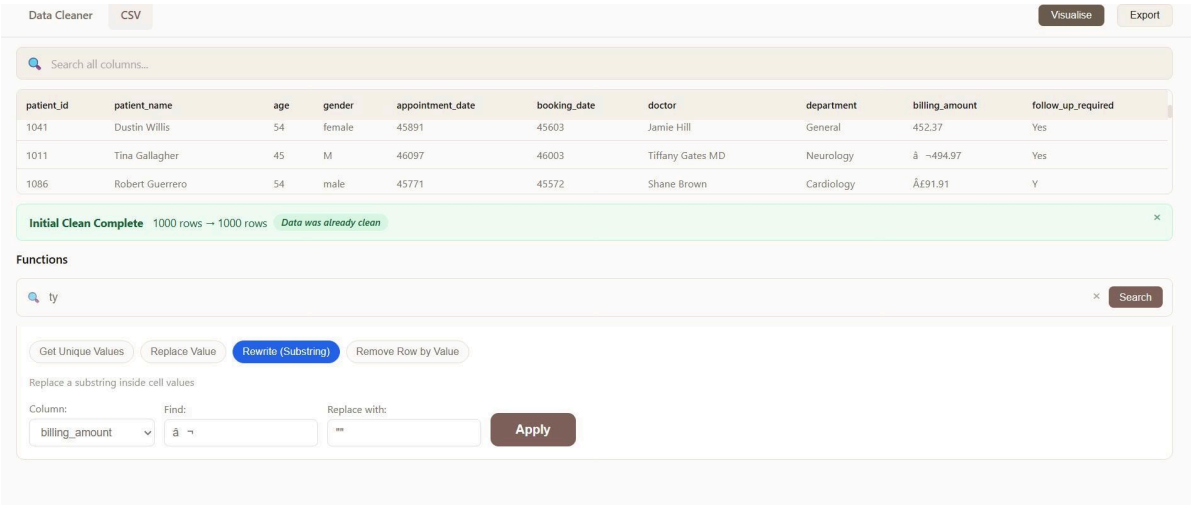
Replace Value can target one exact broken sequence at a time — here, finding 'â ¬3' and mapping it to a placeholder.



Replace Value used to target one specific corrupted sequence in billing_amount.

Second approach: Rewrite (Substring)

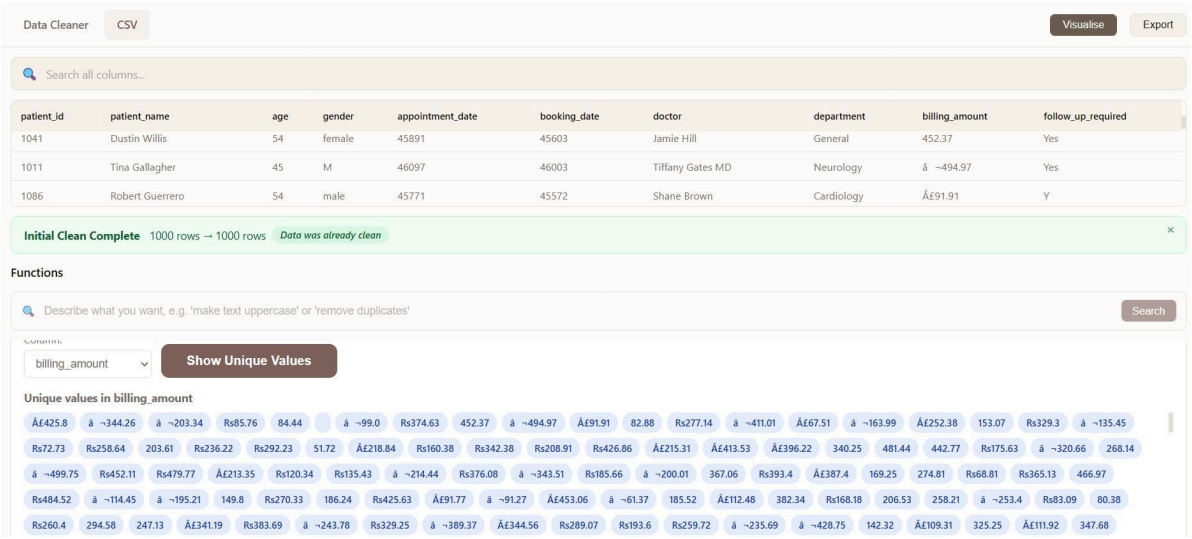
Rewrite (Substring) handles the broader case: find the broken sequence anywhere it appears inside a cell, replace it with the correct character or an empty string, and apply the fix across the whole column in one action — more efficient than fixing one exact value at a time.



Rewrite (Substring) stripping a corrupted currency character from billing_amount across the whole column.

Step 9 — Describe What You Want

Not every user remembers the exact name of a function. The natural language search bar, visible above the function library, lets a user type what they're trying to do in plain English — for example 'make text uppercase' or 'remove duplicates' — and the platform interprets the intent and highlights the matching function, right alongside the same billing_amount data shown a moment ago.

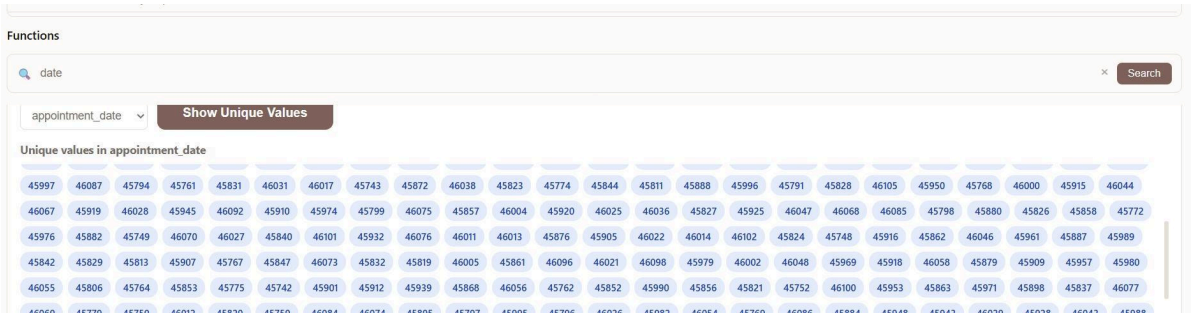


The natural language search bar sits directly above the function library, ready to interpret a plain-English request.

Step 10 — Standardising Excel Date Serial Numbers

Problem

Excel and CSV exports frequently store dates as raw serial numbers rather than readable dates — a value like 45800 actually represents a specific calendar date, counted as days since December 30, 1899. Get Unique Values on appointment_date confirms every value is still in this raw form.



appointment_date entirely in raw Excel serial number form, before standardisation.

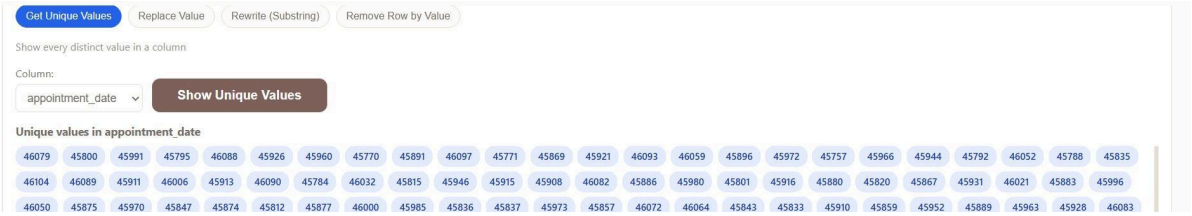
Action

Date Standardize turns any column of these numbers, mixed date strings, or native date objects into one consistent, readable format with a single action.



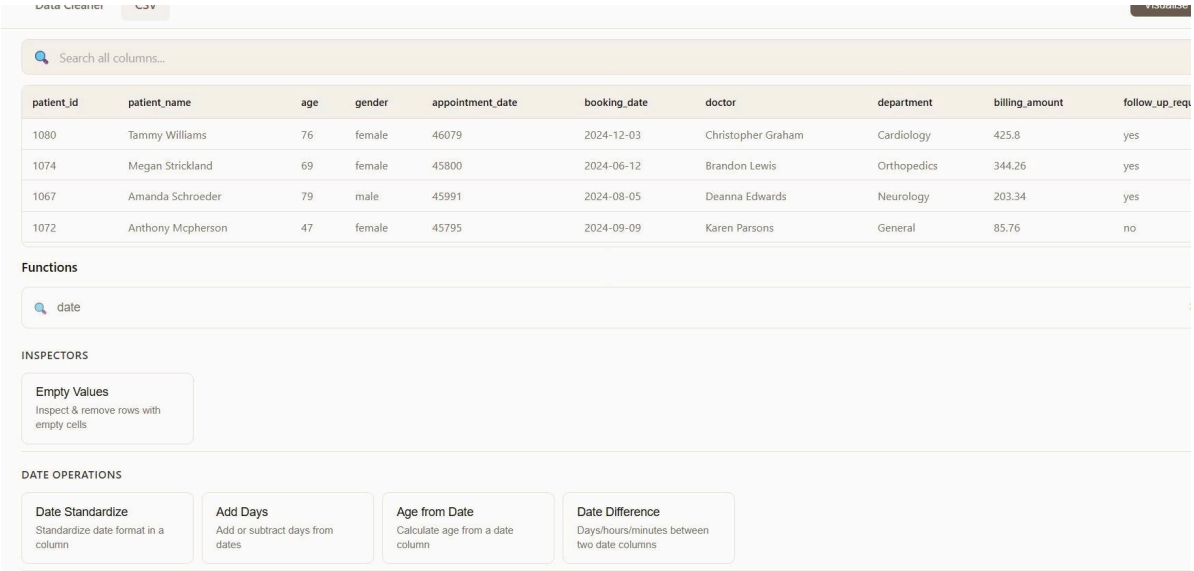
Date Standardize: pick the column and the desired output format, then apply.

The function library groups every date-related tool together, so extracting a year, adding days, calculating age, or standardising format are all one click away from each other.

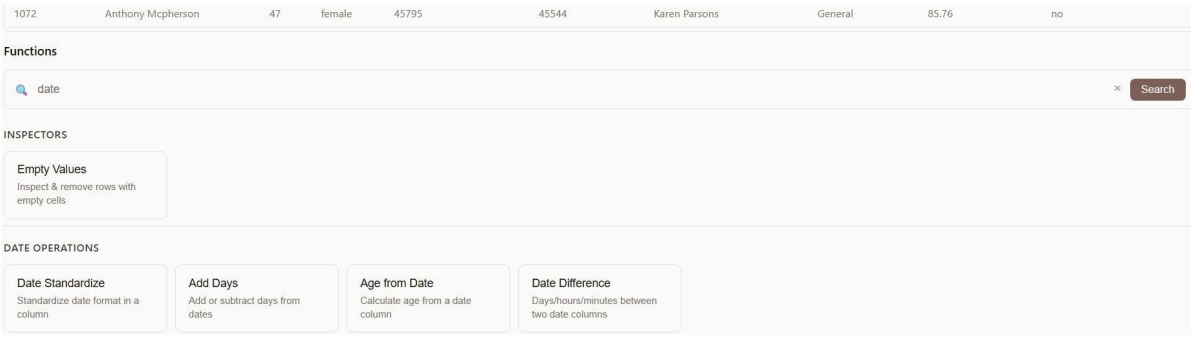


Get Unique Values on appointment_date, showing every value still in its raw Excel serial number form on a later data load.

Applying Date Standardize to booking_date converts the whole column to real, readable dates immediately.

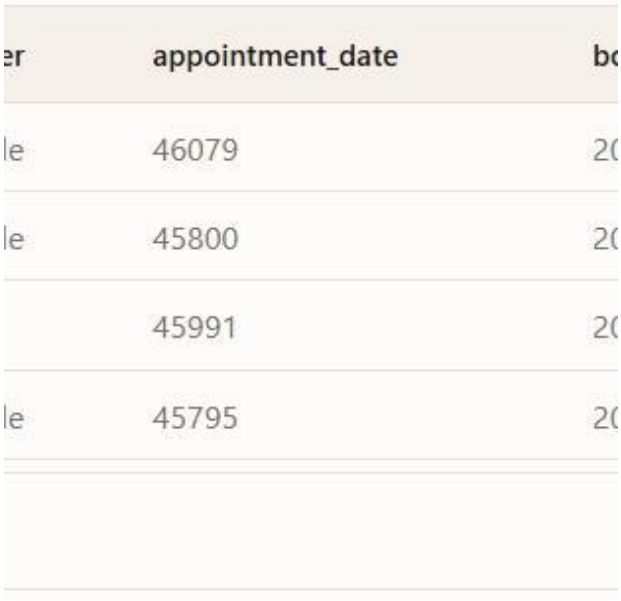


booking_date reads as real dates (2024-12-03, 2024-06-12...) right after running Date Standardize, with the same tool ready to apply to appointment_date next.



The full Date Operations toolset available immediately after standardising a column, ready for the next transformation.

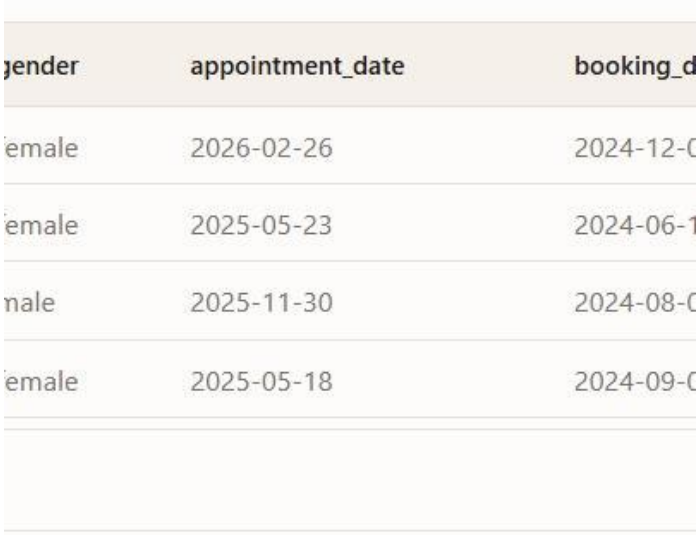
Get Unique Values confirms exactly which columns still need standardising at a glance, so nothing gets missed across a large sheet.



Get Unique Values on appointment_date, confirming it as the next column to standardise.

Result

Running Date Standardize on appointment_date as well brings both columns into a consistent, human-readable format across the whole sheet — the same one-click flow, repeated column by column, however many date fields the sheet contains.



Both appointment_date and booking_date now display as real, readable dates.

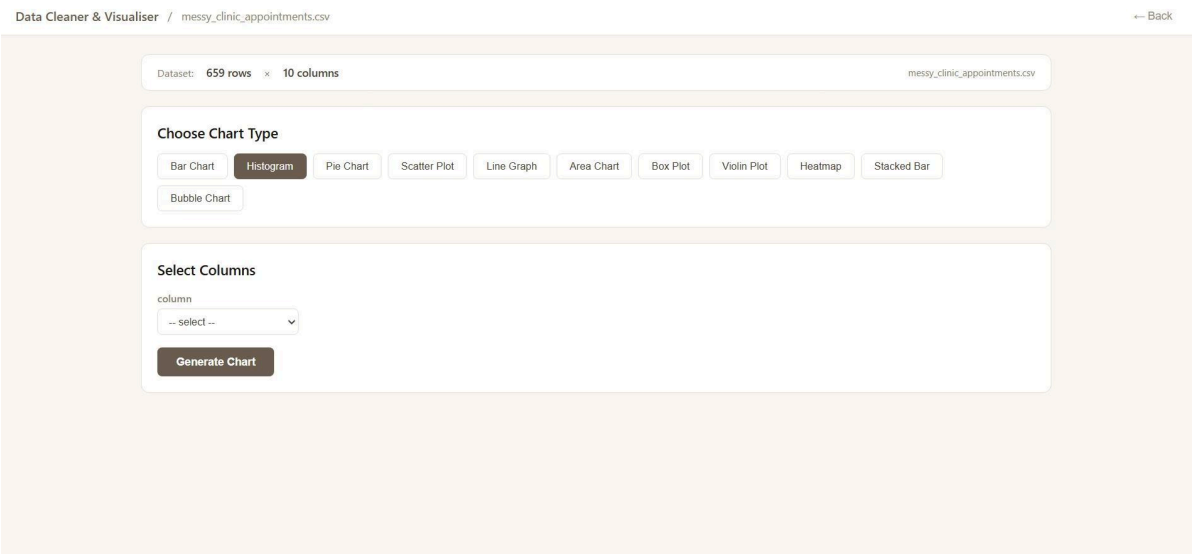
Even the Export and Visualise actions stay one click away throughout the cleaning process, so a fully standardised sheet is always ready to move straight into charts or leave the platform as a finished file.



Visualise and Export stay accessible at every step of the cleaning workflow.

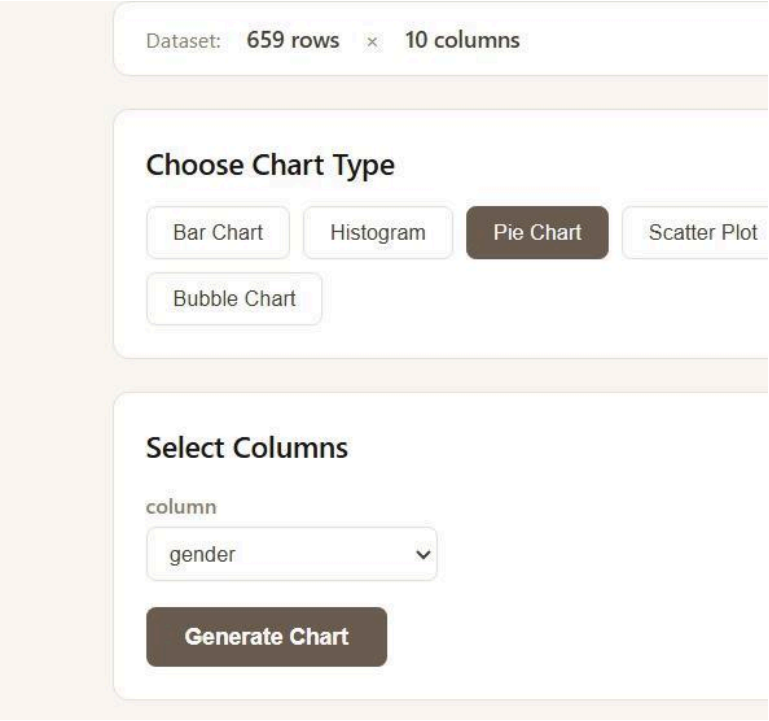
Step 11 — Visualise: AI-Powered Charts

Once a sheet is cleaned, the Visualise screen turns it into charts without leaving the browser. Eleven chart types are available — Bar, Histogram, Pie, Scatter, Line, Area, Box Plot, Violin Plot, Heatmap, Stacked Bar, and Bubble — chosen based on which columns are selected.

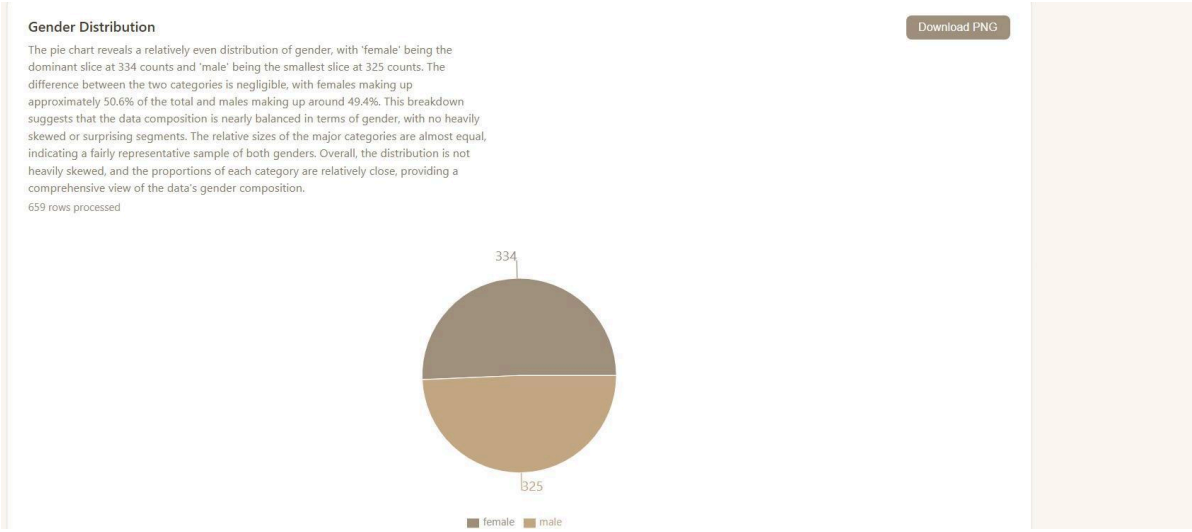


The Visualise screen for messy_clinic_appointments.csv, showing all eleven available chart types.

Selecting Pie Chart and the gender column, then generating, produces a chart along with an AI-written title and a plain-English insight describing what the data actually shows.

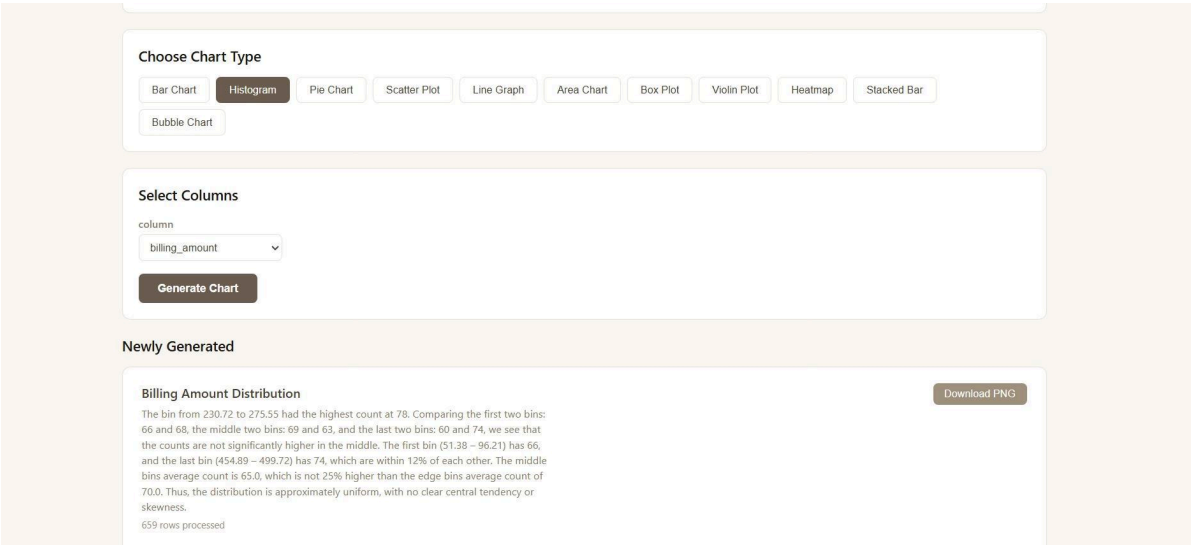


Choosing Pie Chart and the gender column before generating.

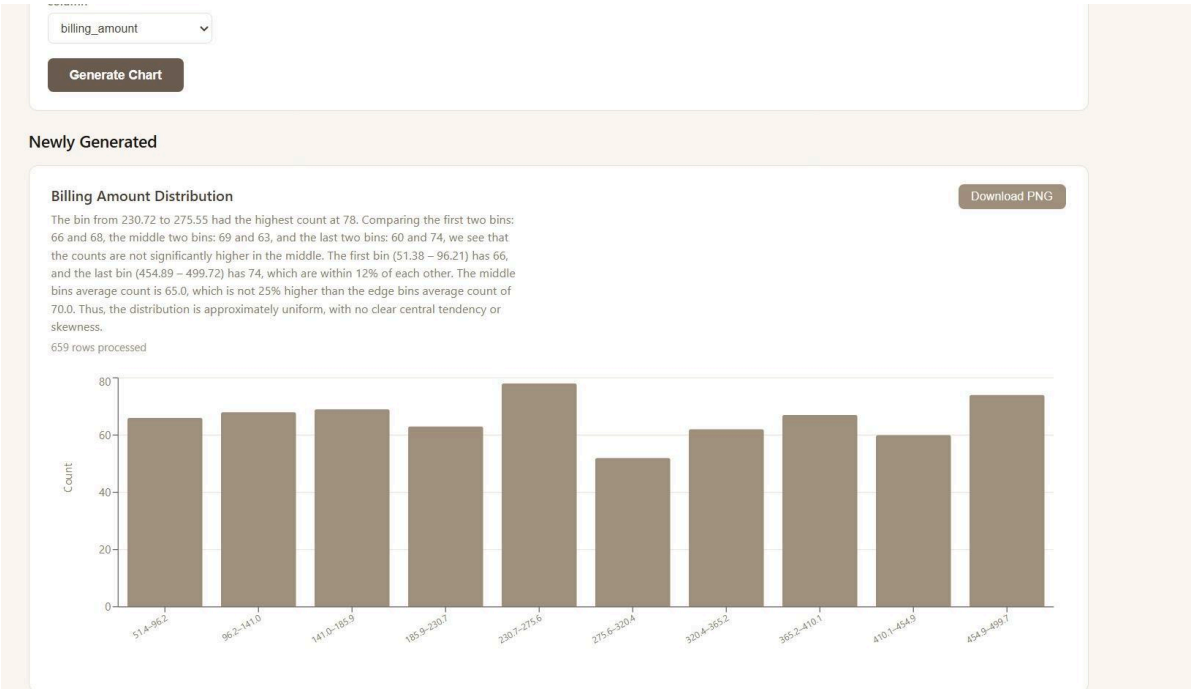


Gender Distribution: a generated pie chart with an AI-written title and insight, computed from the actual counts (334 female, 325 male).

The same flow works for numeric columns. Selecting Histogram on billing_amount generates a distribution chart, again paired with an AI insight that reasons from the actual bin counts rather than assuming a default shape.



Choosing Histogram and the billing_amount column before generating.



Billing Amount Distribution: the AI insight compares edge and middle bins directly and correctly identifies the distribution as approximately uniform.

Summary

- Upload a CSV or Excel file and run a one-click Initial Clean
- Over 40 no-code cleaning functions across ten categories
- Get Unique Values and the Empty Values Inspector surface hidden data quality issues instantly
- Replace Value, Rewrite (Substring), and Regex Replace fix inconsistencies without formulas or code
- Natural language function search for users who don't know the exact function name
- Date Standardize correctly converts Excel serial numbers into readable dates, column by column

- Visualise generates eleven chart types with AI-written titles and insights grounded in the actual computed data
- Export the cleaned sheet at any point

Live deployment: data-cleaner-wine.vercel.app

Source code: github.com/Hlulani-B/Data-Cleaner